25901608 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR/IMPROVEMENT/REAPPEARANCE EXAMINATION APRIL 2026

SECOND SEMESTER

Discipline Specific Core Course (DSC) - MG2DSCBIF100 - BIOLOGICAL DATA AND MANAGEMENT

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short answer Questions

Answer any **5** questions

Each question carries **2** marks

1. Compare cloud databases and on-premises databases in terms of scalability. [U] / [CO1]
2. Articulate two features of Medline. [A] / [CO4]
3. Discuss classification of biological database. [U] / [CO2]
4. Identify scopes of bilogical data. [U] / [CO2]
5. Give one application of motif databases. [A] / [CO4]
6. Establish why GenBank is considered a primary database. [A] / [CO4]
7. Write the purpose of BCDB. [A] / [CO4]

[2x5 = 10]

Part B

Short Essay Questions

Answer any **4** questions

Each question carries **5** marks

8.	Illustrate the flow of data from input to database storage using a suitable diagram and labels.	[U] / [CO1]
9.	Discuss the role of big data in driving decision-making and innovation.	[U] / [CO1]
10.	identify the major databases maintained by NCBI	[U] / [CO2]
11.	Discuss how primary databases differ from secondary databases?	[U] / [CO3]
12.	Articulate the role of the ENZYME database.	[A] / [CO4]
13.	Complete a table listing major genome databases.	[A] / [CO4]

[5x4 = 20]

Part C

Long essay Questions

Answer any **2** questions

Each question carries **10** marks

14.	Compare the main features of SQL, PL/SQL, and T-SQL.	[U] / [CO1]
15.	Compare and contrast the three-tier architecture of a DBMS with the two-tier architecture.	[U] / [CO1]
16.	Describe the key phases in the growth of biological databases.	[U] / [CO2]
17.	Explain why structural databases are useful.	[A] / [CO4]

[10x2 = 20]